

Problem 3

A cube-shaped object with edge length $a = 10$ cm is made of a metal with density $\rho = 8600$ kg/m³. Calculate:

- (a) The mass of the cube on Earth and on the Moon.
 - (b) The weight of the cube on Earth's surface and on the Moon's surface.
- The acceleration due to gravity is $g_z = 9.81$ m/s² on Earth and $g_M = 1.62$ m/s² on the Moon.

Solution

(a) Mass of the cube

The volume of the cube is:

$$V = a^3 = (0.1)^3 = 0.001 \text{ m}^3.$$

The mass of the cube is given by the formula:

$$m = \rho V = 8600 \times 0.001 = 8.6 \text{ kg}.$$

Since mass is independent of location, the mass on the Moon is the same as on Earth:

$$m = 8.6 \text{ kg}.$$

(b) Weight of the cube

Weight is given by:

$$W = mg.$$

On Earth:

$$W_z = 8.6 \times 9.81 = 84.37 \text{ N}.$$

On the Moon:

$$W_M = 8.6 \times 1.62 = 13.93 \text{ N}.$$

Final Answer:

- Mass of the cube: 8.6 kg on both Earth and Moon. - Weight on Earth: 84.37 N. - Weight on the Moon: 13.93 N.